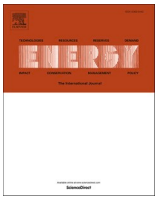
Energy 262 (2023) 125512



Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy





Using energy consumption constraints to control the freight transportation structure in China (2021–2030)

Dajie Zuo [a,*], Qichen Liang [a,b], Shuguang Zhan [c], Wencheng Huang [a], Shenglan Yang [a], Mengyun Wang [a]

[a] School of Transportation and Logistics, Southwest Jiaotong University, Chengdu, 611756, China
[b] Shanghai Municipal Engineering Design Institute (Group) Co., Ltd, Shanghai, 200092, China
[c] School of Automotive and Transportation Engineering, Hefei University of Technology, Hefei, 230009, China

ARTICLE INFO

Keywords:
Freight transportation structure adjustment
Energy consumption
Annual control
FTSA-AC model
Development plan

ABSTRACT

China has formulated a series of energy-saving policies to combat climate change, including energy consumption limit to 6 billion tonnes of standard coal by 2030, thus controlling the energy consumption of freight transportation is crucial for achieving energy-saving goals. In this study, the annual control model for freight transportation structure adjustment (FTSA-AC model) was established from the perspectives of the target year prediction and intermediate year controlling to control energy consumption in the freight sector. Innovatively, an efficient algorithm was designed to solve the model and obtain the annual control plan of freight transportation structure adjustment in China (2021–2030). Accordingly, the national freight transportation structure will be continuously adjusted to the optimal state until 2027, and the freight turnover will be achieved at a steady growth rate of 1.50% from 2027 to 2030. The freight turnover will reach 22888.040 billion tonne-kilometres by 2030, in which railways, highways, waterways, and aviation would account for 18.92%, 30.20%, 50.76%, and 0.12%, respectively. According to the results, freight energy consumption will reach its peak in 2029, with 145 million tonnes of standard coal. Lastly, a series of reasonable policy recommendations were presented to provide a reference for energy-efficient freight transport development in China.

1. Introduction

The shortage of natural resources caused by immoderately energy consumption can trigger serious global energy crises [1]. As the largest developing country around the world, China's industrialisation process has been advancing rapidly, resulting in substantial energy consumption [2]. By 2010, China's total energy consumption had surpassed that of the United States, becoming the largest energy consumer worldwide [3], so China must implement sustainable development principles to balance the environment, economy, and society [4]. To this end, China announced several energy-saving and emission-reduction policies to address issues related to energy consumption.

The transportation sector is a high energy consumer [5]. Recently, transportation energy consumption has accounted for an increasing proportion in China, which has had a significant impact on the national energy conservation efforts [6]. The United States and the United Kingdom successively introduced the National Freight Strategic Plan and the Transport Decarbonisation Plan, respectively, to solve the energy consumption issue in the transport sector. Accordingly, it has been regarded as the most feasible approach to build a green transportation system comprehensively by optimising the transportation structure, vigorously developing rail transportation, and minimising long-distance road transportation of bulk cargo (e.g. coal) [7,8]. In October 2018, China proposed the Three-year Action Plan for Promoting the Adjustment of Transport Structure (2018–2020) to strengthen energy conservation and emission reduction in the transportation sector and clarify the direction of transportation structure adjustment, which required the national railway and waterway freight volume to increase, respectively, by 1.1 billion tonnes and 500 million tonnes — an increment of 30% and 7.5%, respectively, from 2017 to 2020 [9].

Considered the background and regulations of the policies above, the volume of bulk cargo transportation undertaken by railways and waterways, port railway collection and distribution, and container multimodal transport has increased significantly in China. More importantly, breakthroughs have been made in the adjustment of the transportation structures in crucial regions. However, statistics indicate that

* Corresponding author. School of Transportation and Logistics, Southwest Jiaotong University, Chengdu, 611756, China.
E-mail address: zuodajie@swjtu.edu.cn (D. Zuo).

https://doi.org/10.1016/j.energy.2022.125512
Received 21 January 2022; Received in revised form 12 September 2022; Accepted 19 September 2022
Available online 26 September 2022
0360-5442/© 2022 Elsevier Ltd. All rights reserved.

the actual freight volume completed by national railways was 4.46 billion tonnes in 2020, which is still 330 million tonnes less than the freight increase goal of the Three-Year Action Plan. Therefore, exploring a feasible method to formulate a more reasonable annual control plan for freight transportation structural adjustment from 2021 to 2030 is conducive to the formulation of mandatory annual quantitative assessment indicators for the development of the freight transportation sector. In fact, it is of great significance to avoid the occurrence of an energy crisis and to promote the realisation of China's energy-saving and emission reduction goals projected for 2030.

In 2020, the completed freight transportation volume of roadways accounted for 73.8% of the total freight transportation volume in China, and will continue to occupy a dominant position in the comprehensive transportation system. However, there are still many political, economic, and technical obstacles in the process of decarbonising roadway freight transportation [10]. Therefore, by considering the advantages of low energy consumption and low emissions of railways [7,11,12], this study aims to minimise the energy consumption of freight transportation and maximise the increase in the share rate of railway freight turnover. As a requirement of the national total energy consumption control by 2030, the annual control model for freight transportation structure adjustment (FTSA-AC model) was established to achieve the goal of energy consumption control of freight transportation. First, the optimal feasible solution of the freight transportation structure in the target year was obtained by determining the parameters in the model. Second, optimal control of freight transportation structure adjustment in the intermediate year was achieved. Third, the results serve as a useful reference for policymakers in transportation departments to formulate a reasonable annual control plan for freight transportation structure adjustment in the future.

The remainder of this article is organised as follows. Section 2 presents a brief review of energy consumption status of transportation, transportation structure adjustment and the Promotion of energy saving and emission reduction by transportation structure adjustment. Section 3 introduces the basic modelling framework. In Section 4, the FTSA-AC model is established and a solution algorithm is presented. In Section 5, the annual control plan for freight transportation structure adjustment in China (2021–2030) is formulated. Lastly, Section 6 summarises our findings and presents policy recommendations and directions.

2. Literature review

The purpose of this study is to formulate a reasonable annual control plan (2021–2030) for adjusting the freight transportation structure in China based on the priority realisation of the energy consumption control target for freight transportation and to analyse the results of the current status of transportation energy consumption, the direction of transportation structure adjustment, the promotion of energy saving, and emission reduction by considering the transportation structure adjustment in the literature review.

2.1. Energy consumption status of transportation

In the 1970s, by examining the relationship between energy supplies and transportation systems, Goss and Mcgowan pointed out that a major confrontation was forthcoming between energy supplies and transportation in the future [13]. This notion has been confirmed in several countries over recent years. At present, the global transportation energy mix is still dominated by oil derivatives, partially because the technical requirements of this sector makes it difficult to transition to low-carbon energy sources [14]. In 2014, the transport sector accounted for 26% of the global net energy consumption and 64% of the global primary oil consumption was used to satisfy 92% of the total energy demand in the transport sector. In addition to developed countries, with the continuous increase in transportation demand, sizeable growth in transport-related energy consumption is expected in developing countries, such as China, Brazil, India, Indonesia, Mexico, and Turkey [15]. Yu et al. highlighted that transportation is the fastest growing industry in China's energy consumption, and the energy consumption structure of China's transportation sector is dominated by fossil fuels (e.g. diesel, gasoline, kerosene, and fuel oil) [16]. In the U.S., the transportation sector reportedly consumed approximately 28% of the total energy in 2019, making the transportation sector the most significant contributor to anthropogenic greenhouse gas emissions [17].

2.2. Transportation structure adjustment

The goal of transportation structure adjustment is to take advantage of various transport modes to achieve an optimised state with the least energy consumption and highest transportation efficiency. Previously, numerous empirical studies have been dedicated to finding the optimisation direction of transportation structure adjustment from the perspective of the advantages of various transport modes and other influencing factors to achieve the aforementioned goal.

From the perspective of the advantages of various transport modes, Shi et al. established a generalised cost model of railways and highways to analyse the factors that influence the market shares of railways and highways [18]. It pointed out that railways were more competitive in high capacity and long-distance freight transport. Moreover, Xiang et al. proposed the absolute and relative dominant transportation distance models of each freight transport mode on the basis of the mode split rate and provided a quantitative theoretical basis for optimising the transportation structure by exploiting the distinct advantages of various transport modes [19]. Dong et al. analysed the evolution characteristics and trends of China's comprehensive transportation system structure based on theories of information entropy and the leading industry diffusion effect. The outcome indicates that railway would become a stable transport mode that supports the national economy, whereas the highway would focus on concentration and evacuation functions. Meanwhile, the centre of the development of waterways would tilt towards ocean transportation, and aviation would continue to satisfy the transportation demand of high value-added and high time value [20].

Considering other influencing factors, Lee and Yoo analysed the development direction transportation structure in South Korea from an economic perspective using the input-output analysis method and found that South Korea needed to strongly pursue the policy of modal shift of freight transportation from highways to waterways to avoid serious national economic losses [21]. Cai et al. analysed the factors influencing the selection of transport modes from three aspects of demand characteristics, logistics facilities and service level to conclude that Beijing was suitable for adopting the transportation structure adjustment programme of 'highway-to-railway shift', and further summarised several types of cargo and logistics facilities suitable for the programme [22]. Moreover, social emergencies also have a distinct impact on the transportation structure. For instance, China's transportation sector has received multiple shocks of the COVID-19 pandemic from supply and demand sides, particularly the freight transportation using railways, waterways, and aviation, which have been adversely affected and their share of freight turnover has dropped significantly [23].

2.3. Promotion of energy saving and emission reduction by transportation structure adjustment

Carrier reported that energy conservation and emission reduction can be achieved through transportation planning [24]. Thereafter, increasing number of scholars has paid attention to the topic of 'green transportation'. Scholars have regarded energy conservation and emission reduction as two of the most critical goals of transportation structure adjustment.

Furthermore, research on energy conservation and emission reduction in transportation has steadily developed from qualitative induction based on historical analysis and logical reasoning to comprehensive

multidisciplinary quantitative calculations, including statistics, operations research, and so on. Komor conducted a detailed investigation of the relationship between freight transportation and energy consumption in America and clarified the strategic significance of formulating freight transportation policies, such as multi-modal transportation and transportation structure adjustment, to reduce energy consumption [25]. Moreover, Álvarez considered that high-speed railway systems typically consume 29% less energy than conventional railway systems on average, and advocated vigorously developing high-speed railways [26]. Based on the freight turnover and energy consumption of various transport modes in China, Tian et al. calculated the greenhouse gas (GHG) emissions and emission intensity of each freight transportation sector. A comparative analysis of various freight transportation sectors showed that from the perspective of emissions, highways had the largest carbon emissions, accounting for about 85% of the total; from the perspective of emission intensity, highways and aviation had the highest emission intensity. Therefore, it is recommended to improve the energy efficiency of highways and aviation and increase the freight transportation sharing rate of railways, waterways, and pipelines from the transportation structure [27]. Wang et al. used a log-mean divisia index (LMDI) method to decompose the factors that influenced carbon dioxide emissions in China's passenger and freight transportation sectors between 1990 and 2015. According to the results, the transportation structure played a positive role in reducing carbon dioxide emissions for both passenger and freight transportation [28]. Ülengin et al. conducted a fuzzy cognitive map analysis to explore the impact of transportation options on energy consumption, environmental pollution, and climate change. Furthermore, the scenario analysis indicated that clean energy is a better alternative to fossil energy in terms of fuel selection, and the transfer of the share of highways in passenger and freight transportation to railways should be promoted in terms of the transportation structure without damaging the environment [29]. Li and Zhang comprehensively affirmed Ülengin's view through their study on transportation carbon emissions and proposed a series of subsidy policies to encourage the development of railway transportation [30].

On the basis of comprehensively understanding the current status of transportation energy consumption, it is evident that the studies mentioned above have highlighted the direction of transportation structure adjustment comprehensively and have fully affirmed the promotion effect of transportation structure adjustment, particularly the vigorous development of railway transportation on energy saving and emission reduction. To this end, several reasonable quantitative methods have been proposed. However, existing research lacks a comprehensive method for predicting the freight transportation structure adjustment goal in the future and a specific annual control plan. Moreover, China's energy consumption control goal by 2030 is the latest requirement based on basic national conditions and the actual demand for green development. To date, there is no corresponding national-level freight transportation structure adjustment policy formulated to guide local and relevant departments. The annual control plan of freight transportation structure adjustment in China (2021–2030) proposed in this study can make up for this complementary aspect to achieve energy consumption control.

3. Basic modelling framework

Due to the severe impact of COVID-19 on freight transportation by various modes in 2020, 2019 was set as the base year for the study. Combined with the national energy savings for energy consumption control requirements in 2030, 2030 was set as the target year for this study.

Solving the problem of annual control of freight transportation structure adjustment involves reasonably predicting the freight transportation structure in the target year on the basis of priority controlling the energy consumption of freight transportation, and then further realising the control of freight transportation structure adjustment in the intermediate year. The modelling logic is illustrated in Fig. 1.

The following four hypotheses are presented based on the actual situation. (1) All freight transportation is employed using four modes of transport, namely railway, highway, waterway, and aviation. (2) There is a large substitution among various modes of transportation. (3) Both transportation demand and capacity show a continuous and regular growth trend. (4) When modelling, only the energy-saving results brought about by the freight transportation structure adjustment are considered.

The FTSA-AC model was decomposed into two different but coherent models to solve the annual control of freight transportation structure adjustment in China. The details are shown in (i) and (ii).

(i) **Establishment of a prediction model for freight transportation structure adjustment in the target year (FTSA-TP model).** Based on the contradiction between the requirement of energy saving and the reality of a continuous increase in the energy consumption of freight transportation in China, the development of the model comprehensively considers China's fundamental national conditions and the advantages of railway transportation, such as larger transportation capacity, longer transportation distance, and lower energy consumption. The model sets the first goal of minimising the total energy consumption of freight transportation by considering the urgency of alleviating the pressure on the ecological environment caused by the development of transportation and the second goal is to maximise the increase in the share rate of railway freight turnover to obtain a prediction plan for freight transportation structure adjustment in the target year. Meanwhile, since freight transportation structure adjustment is restricted by multiple factors, three principles should be followed. First, the transportation demand brought about by social and economic developments should be satisfied while adjusting the freight transportation structure when developing the model. Second, owing to the constraints of the infrastructure and mobile equipment, there is a maximum carrying capacity for each transport mode. Third, the development of the transportation scale of various transport modes should satisfy the requirements of the energy consumption control of freight transportation.

(ii) **Establishment of a control model for freight transportation structure adjustment in the intermediate year (FTSA-IC model).** Based on a reasonable prediction of the results of the freight transportation structure adjustment in the target year, the relationships among the initial, intermediate, and final states of freight transportation structure adjustment are analysed. A model was developed to formulate the optimal control plan for freight transportation structure adjustment in the intermediate year. It can keep the total freight turnover increasing at a certain speed. Additionally, the growth rate of the freight turnover of various transport modes in the intermediate year was obtained so that the share rate of the freight turnover in the base year reaches the predicted results in the target year. During the development of the model, the following two constraints should be satisfied: The first constraint is that the sum of the product of the share rate of freight turnover and the growth rate of the various transport modes in the intermediate year should be numerically equal to the growth rate of the total freight turnover in the same year. The second constraint is that the growth rate of freight turnover in the intermediate years of various transport modes must satisfy reasonable upper and lower limits.

4. FTSA-AC model

After setting the parameters and variables, the process of dividing the establishment of the FTSA-AC model into two steps is described: the FTSA-TP model and FTSA-IC model. Moreover, an algorithm for solving

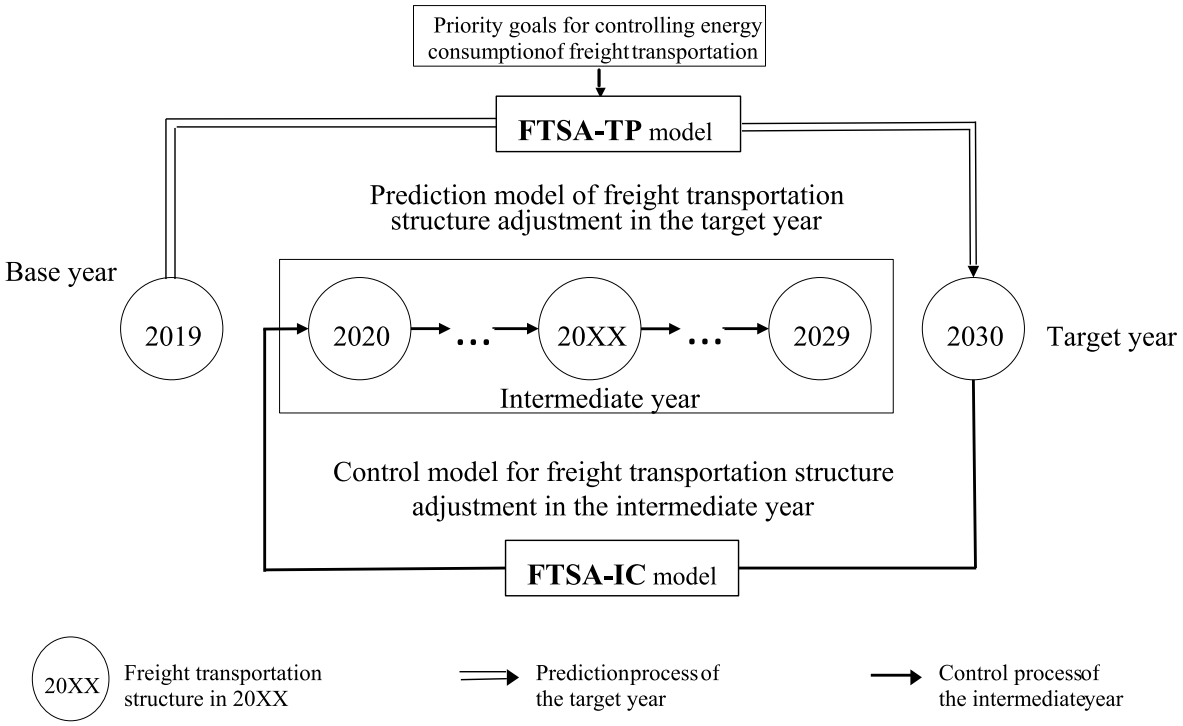


Fig. 1. Modelling logic for annual control of freight transportation structure adjustment.

the FTSA-AC model is provided.

4.1. Description of main parameters and variables

The developed model has the following parameters and variables in this study:

Y	Goal of the freight transportation structure adjustment
Z	Control plan of the freight transportation structure adjustment in the intermediate year
$E(0)$	Total energy consumption of freight transportation in the base year
$E(n)$	Expected value of the total energy consumption of freight transportation in the target year
ΔQ	Expected value of the increase in the share rate of railway freight turnover
$Q_i(0)$	Freight turnover of transport mode i in the base year
$Q_i(k)$	Freight turnover of transport mode i in year k
$Q_i(n)$	Freight turnover of transport mode i in the target year
$Q_i^{\text{demand}}(n)$	Freight transportation demand of transport mode i in the target year
$Q_i^{\text{capacity}}(n)$	Freight transportation capacity of transport mode i in the target year
$Q_i^{\text{lower}}(n)$	Lower limit of the freight transportation development scale of transport mode i in the target year
$Q_i^{\text{upper}}(n)$	Upper limit of the freight transportation development scale of transport mode i in the target year
$e_i(n)$	Energy consumption coefficient of transport mode i in the target year
$x_i(0)$	Freight turnover sharing rate of transport mode i in the base year
$x_i(k)$	Freight turnover sharing rate of transport mode i in year k
$x_i(n)$	Freight turnover sharing rate of transport mode i in the target year
$\mathbf{X}(n)$	Target-vector of ideal freight transportation structure
$\mathbf{X}(k)$	Vector of freight transportation structure in year k
$u_i(k)$	Growth rate of freight turnover of transport mode i in year k+1 compared to year k
$\mathbf{U}(k)$	Vector of growth rate of freight turnover in year k+1 compared to year k
v	Annual growth rate of total freight turnover
m_i	Lower limit of the annual freight turnover growth rate of transport mode i
M_i	Upper limit of the annual freight turnover growth rate of transport mode i
c_i	Lower limit of the share rate of freight turnover of transport mode i
C_i	Upper limit of the share rate of freight turnover of transport mode i
i	$i = 1, 2, \cdots, r(r = 4)$, respectively representing the four transport modes: railway, highway, waterway, and aviation

4.2. FTSA-TP model

4.2.1. Objectives

The first objective is to minimise the total energy consumption of freight transportation, as shown in Eq. (1):

$$\min \left[\sum_{i=1}^{r} Q_i(n) e_i(n) \right] \tag{1}$$

where the total energy consumption of freight transportation is equal to the sum of the product of the freight turnover of various transport modes and their energy consumption coefficients in the target year.

The second objective is to maximise the increase in the share rate of railway freight turnover, as shown in Eq. (2).

$$\max \left[Q_1(n) \Big/ \sum_{i=1}^{r} Q_i(n) - Q_1(0) \Big/ \sum_{i=1}^{r} Q_i(0) \right] \tag{2}$$

where the ratio of railway freight turnover to total freight turnover in the base year and target year are calculated, and their difference is used to express the increase in the share rate of railway freight turnover.

4.2.2. Constraints

According to the three principles in the process of modelling proposed in Section 3, three types of constraints were established to determine the feasible region and optimal solution of the objective function in this study.

(i) The first constraint originates from economic and social development, which ensures that the total transportation supply satisfies the total transportation demand in the target year:

$$\sum_{i=1}^{r} Q_i(n) \geq \sum_{i=1}^{r} Q_i^{\text{demand}}(n) \tag{3}$$

(ii) The second constraint arises from restrictive conditions for the development of various transportation modes. The freight

turnover of any transport mode in the target year should satisfy the scope of its development scale limitation in the target year; meanwhile, it cannot exceed the transportation capacity:

$$Q_i^{\text{lower}}(n) \le Q_i(n) \le Q_i^{\text{upper}} \le Q_i^{\text{capacity}} \tag{4}$$

(iii) The third constraint arises from the requirement for vigorously developing railway freight transportation. While increasing the share rate of railway freight turnover, its absolute value of railway freight turnover should be improved.

$$Q_1(n) - Q_1(0) \ge 0 \tag{5}$$

4.2.3. Model composition

(1) Based on the objectives and constraints above, the multi-objective planning model is developed as follows.

formula(1)

formula(2)

$$\text{s.t.}\begin{cases} \text{formula}(3) - (5) \\ Q_i(n) \ge 0, i = 1, 2, \cdots, r \end{cases} \tag{6}$$

(2) In the process of formulating mid-to long-term development planning for more than a decade, a certain degree of flexibility should be set for objectives. Therefore, the bi-objective nonlinear programming problem proposed in this study was converted to the FTSA-TP model with positive and negative deviations, and this model is a Goal Planning model. The objective functions Eq. (1)(2) were transformed into constraints with positive and negative bias variables in the model. The composition of the model is shown in Eq. (7):

$$\min Y = P_1 d_1^+ + P_2 d_2^-$$

$$\text{s.t.}\begin{cases} \sum_{i=1}^{r} Q_i(n) e_i(n) \Big/ E(0) - d_1^+ + d_1^- = E(n) \Big/ E(0) \\ \left[Q_1(n) \Big/ \sum_{i=1}^{r} Q_i(n) - Q_1(0) \Big/ \sum_{i=1}^{r} Q_i(0) \right] - d_2^+ + d_2^- = \Delta Q \\ \text{formula}(3) - (5) \\ d_1^+, d_1^-, d_2^+, d_2^- \ge 0, i = 1, 2, \cdots, r \end{cases} \tag{7}$$

The FTSA-TP model is decomposed into two-level models: LP_1 and LP_2. where P_1 and P_2 denote the first-and second-priority factors of the model, respectively ($P_1 \gg P_2$). In order to make the dimensions of variables consistent, energy goal is expressed as minimising the ratio of freight energy consumption in the target year to the base year. Meanwhile, $d_1^{\pm}$ and $d_2^{\pm}$ represent the positive and negative deviation variables of the optimal solution of the model relative to $E(n)/E(0)$ and ΔQ, respectively. To pursue the goal of freight transportation structure adjustment in this study, d_1^+ and d_2^- are selected respectively to represent the pursuit of making the total energy consumption of freight transportation as low as possible and the increment in the share rate of railway freight turnover as high as possible compared with their expected value.

4.3. FTSA-IC model

Based on the statistical data and the prediction results in Section 4.2, it can be calculated that the share rates of freight turnover of transport mode i in the base year and target year are $x_i(0) = Q_i(0) / \sum_{i=1}^{r} Q_i(0)$ and $x_i(n) = Q_i(n) / \sum_{i=1}^{r} Q_i(n)$, respectively. Furthermore, the target-vector of the ideal freight transportation structure can be expressed as $\mathbf{X}(n) = [x_1(n), x_2(n), \cdots, x_r(n)]^{\mathrm{T}}$.

4.3.1. State equation

Set the vector of freight transportation structure in year k as $\mathbf{X}(k) = [x_1(k), x_2(k), \cdots, x_r(k)]^{\mathrm{T}}$, and the vector of the growth rate of freight turnover in year k+1 compared to year k as $\mathbf{U}(k) = [u_1(k), u_2(k), \cdots, u_r(k)]^{\mathrm{T}}$. List the state equations as shown in Eqs. (8)-(10):

State of the base year: $\sum_{i=1}^{r} x_i(0) = 1$ (8)

State of the intermediate year: $x_i(k+1) = \frac{x_i(k)[1+u_i(k)]}{1+v}$ (9)

State of the target year:

$$x_i(n) = \frac{x_i(n-1)[1 + u_i(n-1)]}{1 + v} \tag{10}$$

4.3.2. Constraints

According to the description in Section 3, $u_i(k)$ must satisfy the two conditions in Eq. (11):

$$\begin{cases} \sum_{i=1}^{r} x_i(k) u_i(k) = v \\ m_i \le u_i(k) \le M_i \end{cases} \tag{11}$$

4.3.3. Model composition

In this study, the problem of selecting an optimal control plan in the intermediate year which traverses the state of the intermediate year $x_i(k)$ and reaches the state of the target year $x_i(n)$ from the state of the base year $x_i(0)$ must be solved. The problem is converted into the FTSA-IC model, as shown in Eq. (12):

$$\min Z = \sum_{k=0}^{n-1} \sum_{i=1}^{r} \left\{ \frac{x_i(k)[1 + u_i(k)]}{1 + v} - x_i(n) \right\}^2$$

$$\text{s.t.}\begin{cases} \text{formula } (8) - (11) \\ i = 1, 2, \cdots, r, k = 0, 1, 2, \cdots, n-1 \end{cases} \tag{12}$$

To solve the model more conveniently, Eq. (12) can be rewritten in the form shown in Eq. (13), referring to the research results of Tang et al. [31]:

$$\min f[u_i(k)] = \sum_{i=1}^{r} \left\{ \frac{x_i(k)[1 + u_i(k)]}{1 + v} - x_i(n) \right\}^2$$

$$\text{s.t.}\begin{cases} g_j[\mathbf{U}(k)] = \sum_{i=1}^{r} x_i(k) u_i(k) - v = 0, j = 0 \\ g_j[\mathbf{U}(k)] = u_i(k) - m_i \ge 0, j = 1, 2, \cdots, r \\ g_j[\mathbf{U}(k)] = M_i - u_i(k) \ge 0, j = r+1, r+2, \cdots, 2r \\ i = 1, 2, \cdots, r, k = 0, 1, 2, \cdots, n-1 \end{cases} \tag{13}$$

4.4. Model solution algorithm

The FTSA-AC model is a comprehensive model that solves the problem of a bi-objective program and the selection of an optimal control plan for intermediate years. The analytic hierarchy process (AHP) can be integrated to solve multi-objective programming problems and Zoutendijk's feasible direction method is suitable for constrained optimisation problems [32,33]. Moreover, a solution algorithm called analytic hierarchy-feasible direction was designed in this study. The algorithm can optimise the two-level goal of minimising the total energy consumption of freight transportation and maximising the increase in the share rate of railway freight turnover to obtain the optimal control plan for freight transportation structure adjustment in the intermediate year. The procedure for the algorithm is illustrated in Fig. 2. In the algorithm, D and λ are the feasible descent direction and the step length, respectively.

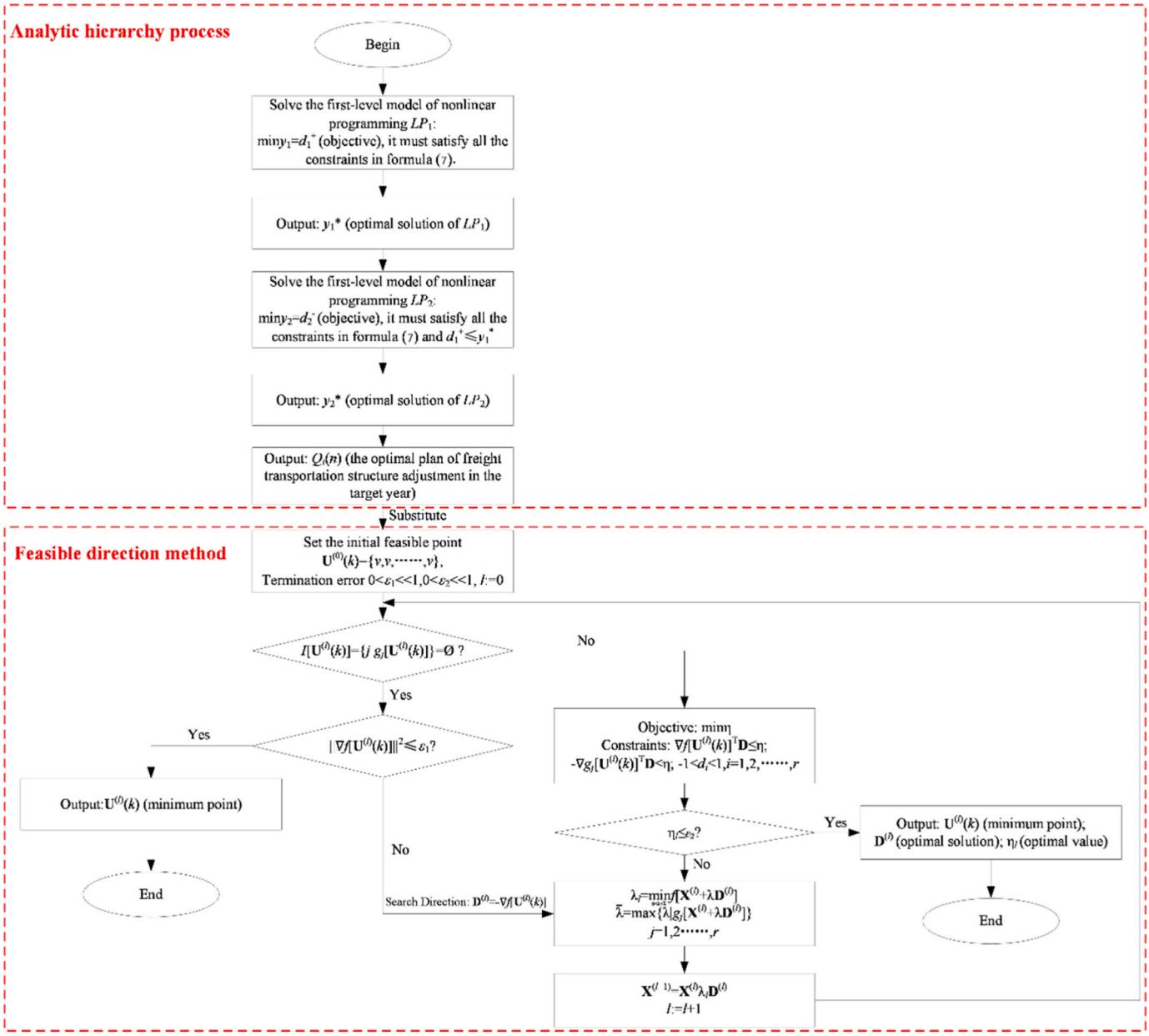


Fig. 2. Procedure of algorithm.

5. Annual control plan of freight transportation structure adjustment in China (2021–2030)

As stated above, the base year and target year were set as 2019 and 2030, respectively, and the value of n was 11. The annual control plan of freight transportation structure adjustment in China (2021–2030) was formulated by solving the FTSA-TP and FTSA-IC models.

5.1. Parameters determination

It was necessary to determine the unknown parameters in the model prior to the model solution. Owing to the differences in the degree of social and economic development in different periods, remote historical data cannot be utilized as a basis for future predictions. Therefore, statistical data from 2011 to 2019 were selected to determine the parameters (the description of the original data is shown in Appendix A, and the parameter determination method is shown in Appendix B). Moreover, the results of the parameter determination are listed in Table 1.

5.2. Model solution results

The parameter determination results in Section 5.1 were adopted to solve the FTSA-AC model developed in this study.

(i) Freight transportation structure in the target year

The freight structure adjustment target that minimises the total energy consumption of freight transportation and maximises the increase in the share rate of railway freight turnover in 2030 was obtained by solving the FTSA-TP model in Section 4.2 (as shown in Table 2). The total energy consumption of freight transportation is 144.9 million tonnes of standard coal by 2030. Furthermore, the share rate of freight turnover for various transport modes in the target year (2030) can be calculated. Fig. 3 compares the share rate of freight turnover in the base year (2019) and the target year (2030).

The FTSA-IC model was solved based on the freight transportation structure in the target year. As shown in Table 3, the optimal control

Table 1
Parameters determination results.

(a)				
Freight turnover in the base year (100 million tonne-kilometres)	$Q_1(0)$	30181.95		
	$Q_2(0)$	59636.39		
	$Q_3(0)$	103963.04		
	$Q_4(0)$	263.20		
	$\sum_{i=1}^{4} Q_i(0)$	194044.58		
(b)				
Freight transportation demand in the target year (100 million tonne-kilometres)	$Q_1^{\text{demand}}(n)$	34349.36		
	$Q_2^{\text{demand}}(n)$	70014.54		
	$Q_3^{\text{demand}}(n)$	111807.88		
	$Q_4^{\text{demand}}(n)$	285.19		
	$\sum_{i=1}^{4} Q_i^{\text{demand}}(n)$	216465.97		
(c)				
Freight transportation capacity in the target year (100 million tonne-kilometres)	$Q_1^{\text{capacity}}(n)$	43582.63		
	$Q_2^{\text{capacity}}(n)$	76839.50		
	$Q_3^{\text{capacity}}(n)$	119043.78		
	$Q_4^{\text{capacity}}(n)$	281.39		
(d)				
Energy consumption coefficient in the target year (Tonnes of standard coal per 100 million tonne-kilometres)	$e_1(n)$	328.00		
	$e_2(n)$	1540.00		
	$e_3(n)$	199.40		
	$e_4(n)$	3903.40		
(e)				
Range of annual growth rate of freight turnover	m_1	0.20%	M_1	5.80%
	m_2	0.40%	M_2	6.70%
	m_3	0.40%	M_3	7.10%
	m_4	0.10%	M_4	7.80%
(f)				
Regulation range of the share rate of freight turnover	c_1	15.55%	C_1	20.00%
	c_2	30.00%	C_2	36.00%
	c_3	50.00%	C_3	58.00%
	c_4	0.12%	C_4	0.15%
(g)				
Freight transportation development scale (100 million tonne-kilometres)	$Q_1^{\text{lower}}(n)$	33659.06	$Q_1^{\text{upper}}(n)$	43291.39
	$Q_2^{\text{lower}}(n)$	64937.09	$Q_2^{\text{upper}}(n)$	77924.51
	$Q_3^{\text{lower}}(n)$	108228.49	$Q_3^{\text{upper}}(n)$	125545.04
	$Q_4^{\text{lower}}(n)$	259.74	$Q_4^{\text{upper}}(n)$	324.69
(h)				
Miscellaneous parameters	v			1.5%
	$E(0)$(100 million tonnes of standard coal)			1.443
	$E(n)$(100 million tonnes of standard coal)			1.450
	ΔQ			5.00%

Table 2
Freight turnover volume of various transport modes in the target year (2030).

transport modes	$Q_i(n)$ (100 million tonne-kilometres)
Railway	43291.39
Highway	69126.39
Waterway	116181.23
Aviation	281.39
Total	228880.40

plan for the freight transportation structure adjustment in the intermediate year can be obtained. The annual control of freight turnover for various transport modes from 2019 to 2030 is shown in Fig. 4. We can delineate the trend of the total energy consumption control goal of freight transportation in these years by combining the prediction results in energy consumption coefficients of various transport modes through further calculations, as shown in Fig. 5.

5.3. Results analysis

From the perspective of freight transportation sector development. With the continuous and steady growth of China's economy, market demand for freight transportation is increasing. Therefore, it is necessary to incessantly improve the freight transportation capacities of various transport modes by expanding the scale of infrastructure construction and increasing the volume of mobile equipment. It is expected that by 2030, the freight transportation capacity of railways, highways, waterways, and aviation will increase by 39.19%, 34.42%, 11.96%, and 5.39%, respectively, compared to 2019, all of which can satisfy market demand.

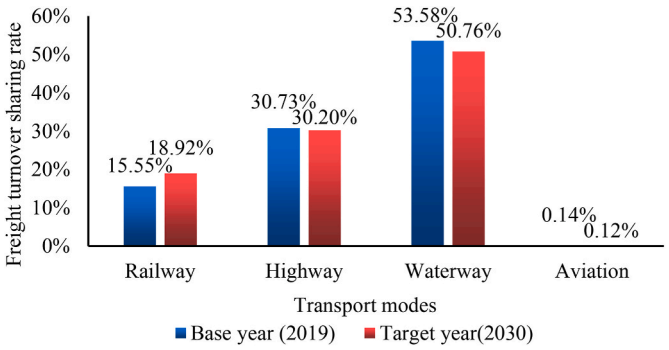


Fig. 3. Comparison of freight transportation structure between base year and target year.

(ii) Optimal control plan of freight transportation structure adjustment in the intermediate year

From the perspective of energy consumption control. In the context of increasing freight transportation volume of various transport modes, relevant departments should take measures to reduce the total energy consumption of freight transportation. These measures must include strengthening transportation organisations and improving energy saving technologies to reduce the energy consumption coefficient. It is expected that by 2030, the energy consumption coefficients of railways, highways, waterways, and aviation will decrease by 16.75%, 9.41%, 23.07%, and 4.13%, respectively, compared with 2019. On this basis, freight transportation energy consumption will reach a peak of 145.0 million tonnes of standard coal by around 2029 and will fall to 144.9 million tonnes of standard coal by 2030, basically achieving national energy consumption control.

6. Conclusion

In this study, the FTSA-AC model was established to control the energy consumption of the freight transportation sector and promote the realisation of national restrictive targets of energy-saving and emission reduction. The model includes two parts: target year prediction and intermediate year control of freight transportation structure adjustment. Moreover, the annual control plan of freight transportation structure adjustment in China (2021–2030) was formulated through an algorithm designed to solve the above model.

6.1. The main findings of this work

Based on the above study, the annual control plan for freight transportation structure adjustment in China (2021–2030) is formulated as follows:

(i) **Freight turnover:** By 2030, the total freight turnover will reach 22888.040 billion tonne-kilometres in China, in which the freight turnover of railways, highways, waterways, and aviation will reach 4329.139 billion tonne-kilometres, 6912.639 billion tonne-kilometres, 11618.123 billion tonne-kilometres, and 28.139 billion tonne-kilometres, respectively (Table 2). Compared to 2019, the freight turnover of the respective transport modes increased by approximately 43.43%, 15.91%, 11.75%, and

Table 3
Optimal control plan of freight transportation structure adjustment in the intermediate year.

	Railway		Highway		Waterway		Aviation	
	$u_1(k)$	$x_1(k)$	$u_2(k)$	$x_2(k)$	$u_3(k)$	$x_3(k)$	$u_4(k)$	$x_4(k)$
2019	–	0.1555	–	0.3073	–	0.5358	–	0.0014
2020	0.0580	0.1619	0.0123	0.3061	0.0065	0.5306	0.0006	0.0015
2021	0.0522	0.1678	0.0130	0.3055	0.0100	0.5253	0.0012	0.0013
2022	0.0478	0.1733	0.0138	0.3051	0.0105	0.5203	0.0010	0.0013
2023	0.0433	0.1781	0.0131	0.3046	0.0092	0.5160	0.0010	0.0013
2024	0.0390	0.1823	0.0124	0.3038	0.006	0.5126	0.0011	0.0013
2025	0.0357	0.1860	0.0119	0.3029	0.0016	0.5099	0.0012	0.0012
2026	0.0320	0.1892	0.0122	0.3020	0.0170	0.5076	0.0011	0.0012
2027	0.0150	0.1892	0.0150	0.3020	0.0150	0.5076	0.0150	0.0012
2028	0.0150	0.1892	0.0150	0.3020	0.0150	0.5076	0.0150	0.0012
2029	0.0150	0.1892	0.0150	0.3020	0.0150	0.5076	0.0150	0.0012
2030	0.0150	0.1892	0.0150	0.3020	0.0150	0.5076	0.0150	0.0012

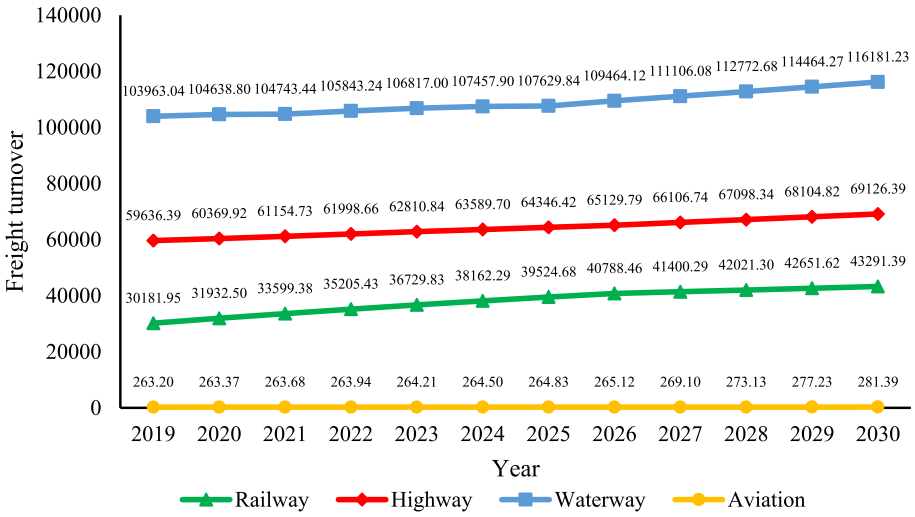


Fig. 4. Annual control of freight turnover of various transport modes from 2019 to 2030 (100 million tonne-kilometres).

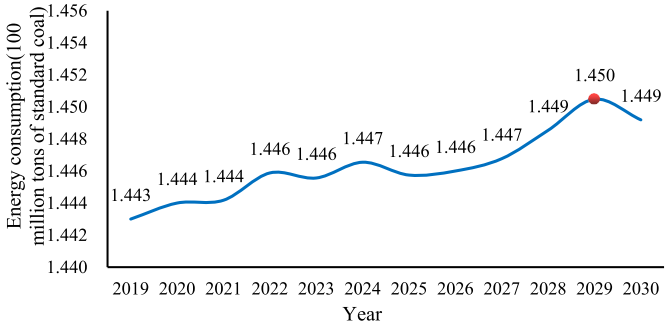


Fig. 5. Trend of total energy consumption control goal of freight transportation from 2019 to 2030 (100 million tonnes of standard coal).

6.91%, respectively, with average annual growth rates of 3.95%, 1.45%, 1.07%, and 0.63%, respectively (Fig. 4).

(ii) **Share rate of freight turnover:** By 2030, the share rate of railway freight turnover will reach 18.92%, an increase of approximately 3.37% from 2019. In contrast, compared with 2019, the share rate of freight turnover on highways, waterways, and aviation decreased by 0.35%, 1.77%, and 0.01%, respectively (Table 3).

(iii) **Freight transportation structure:** From 2021 to 2026, the freight structure will continue to be adjusted and optimised, and it will be in an optimal state around 2027. The freight turnover of various transport modes will achieve steady growth at an annual rate of 1.50% from 2027 to 2030 (Table 3).

(iv) **Energy consumption control:** By 2030, the energy consumption coefficients of railways, highways, waterways, and aviation will decrease by 16.75%, 9.41%, 23.07%, and 4.13%, respectively, compared with 2019. During this process, Freight transportation energy consumption will reach a peak of 145.0 million tonnes of standard coal by around 2029 and will fall to 144.9 million tonnes of standard coal by 2030(Fig. 5).

6.2. Policy recommendations

Based on the above results, the following policy recommendations for freight transportation structure adjustment are proposed to reduce the energy consumption of freight transportation:

(i) **Bring the government's macro-control function into play sufficiently.** Due to management problems, different transportation departments in China have formulated distinct development plans, which lacks effective connections and introduces mutually exclusive market competition. It is necessary to formulate a scientific and reasonable quantitative freight development comprehensive plan by strengthening the top-level design by referring to the annual control plan of freight transportation structure adjustment in China (2021–2030) proposed in this study (shown in Section 5.3). U.S. Department of Transportation has National Freight Strategic Plan to define strategies to achieve transport goals. China should also have a similar

document to plan inter-modal transport through higher authorities to enhance coordination between the various modes of transport.

(ii) **Reasonably control the expansion of transportation infrastructure.** It is suggested that measures such as controlling investment in infrastructure construction and focusing on improving the quality and efficiency of the comprehensive transportation network should be adopted to reasonably control the expansion and expansion of infrastructure to avoid unnecessary waste of transportation capacity. By 2030, the operating mileage of railways (excluding high-speed railways), highway mileage and civil aviation route mileage will reach 0.1221, 5.2369, and 11.8348 million kilometers respectively, and the number of berths in major ports above designated size will reach 15470, which is sufficient (shown in Appendix B3).

(iii) **Promote the process of 'highway-to-railway shift'.** The vigorous development of railway transportation is of great significance to controlling energy consumption, and railways have irreplaceable advantages in bulk cargo and mid-long distance freight transportation. Railway is the dominant transportation mode used for shipments that move over 1200 km [7]. Statistics indicate that the average distance of China's railway freight transportation in 2019 was 688 km. Indeed, there is a great disparity in the use of railway freight transportation between China and the United States. Generally, measures should be taken to accelerate the construction of 'highway-to-railway shift' projects, such as opening up railway sidings for logistics parks or industrial enterprises, to make full use of railway freight transportation capacity in China.

(iv) **Encouraging the development of modern multi-modal transportation.** Compared with the single transport mode of highways, multi-modal transportation can increase transportation efficiency by 30%, reduce freight damage by 10%, reduce transportation costs by 20%, reduce highway congestion by 50%, and promote energy savings by 1/3 [34]. As an effective means to improve the enthusiasm of the freight transportation sector for the development of multi-modal transportation, China should establish a reasonable subsidy programme to provide economic compensation, such as tax abatement and charge reduction, for the departments of multi-modal transportation equipment production and operation.

(v) **Promote the development and application of energy-saving transportation equipment.** While promoting the freight structure adjustment, the development and application of energy-saving transportation equipment should also be encouraged. Using certain technical means, reduce the energy consumption coefficients of railways, highways, waterways and civil aviation to 328, 1540, 199.4 and 3903.4 tonnes of standard coal per 100 million tonne-kilometres respectively by 2030 (shown in Appendix B5). Specific measures include: promoting the application of electric or hybrid electric vehicles, increasing the proportion of electrified railways, and improving the traction efficiency of railway locomotives, etc.

(vi) **Intensify the assessment of relevant localities and departments.** It is necessary to improve the annual work assessment mechanism for the adjustment of the freight structure from 2021 to 2030 based on the annual control plan for freight structure adjustment (2021–2030). We propose to set the average annual growth rates of railway, highway, waterway and civil aviation cargo turnover at 3.95%, 1.45%, 1.07% and 0.76%, respectively. It is required that the sharing rate of railway freight turnover should reach more than 18.91% by 2030. Two time points should be focused on, that is, whether the freight structure will reach the best state around 2027 (the average annual growth rate will be stable at more than 1.5%), and whether the total freight energy consumption will reach the expected peak in 2029 (shown in Section 5.3 and 6.1).

6.3. Outlook

There are still other issues to be further explored. First, the optimisation degree of the model developed in this study can be systematically evaluated and perfected by referring to the historical data of other countries or regions to obtain a more accurate annual control plan for freight transportation structure adjustment. Second, the determination of parameters in the model can be optimised on the basis of considering the current development trends of energy saving technologies, such as the design of new energy vehicles and the "electricity instead of oil" of railway locomotives. Finally, combined with the results of this study, a reasonable annual control plan for the development of transportation infrastructure and mobile equipment can be formulated, which is compatible with the adjustment of the freight transportation structure.

Credit author statement

Dajie Zuo: Project administration, Conceptualisation, Methodology, Funding acquisition and Supervision. **Qichen Liang**: Methodology, Software, Writing – original draft, Writing-review and editing. **Shuguang Zhan**: Writing-review. Wencheng Huang: Writing-review. **Shenglan Yang**: Software and Formal analysis. **Mengyun Wang**: Software analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study was supported by The National Social Science Foundation of China (No. 15BGL003) and The Research Project of the National Railway Administration of China (No. KF2021-29). The projects are mainly dedicated to the study of the evolution and optimisation of the integrated transportation system under low carbon constraints. The School of Transportation and Logistics, Key Laboratory of Comprehensive Transportation of Sichuan Province, National and Local Joint Engineering Laboratory of Comprehensive Intelligent Transportation, and National Engineering Laboratory of Integrated Transportation Big Data Application Technology provide a valuable research environment and communication platform for this study. The authors would like to thank Ms. Rong Huang from Shenzhen SF Taisen Holdings (Groups) Inc. and Ms. Zhan Cao from the Institute of Transportation Development Strategy & Planning of Sichuan Province because their study foundation on related issues provides the possibility for the realisation of this study. Finally, the authors appreciate the helpful comments and recommendations of the anonymous reviewers and editor of this journal.

Appendix

Appendix A: Description of original data

The original data for this research were sourced from the Statistical Bulletin of National Economy and Social Development, the Railway Statistical Bulletin, the Statistical Bulletin of Transportation Industry Development, and the Civil Aviation Statistical Bulletin of Transportation Industry Development.

Table A-1
Freight turnover of various transport modes from 2011 to 2019 (100 million tonne-kilometres)

	2011	2012	2013	2014	2015	2016	2017	2018	2019
Railway	29465.79	29187.09	29173.89	27530.19	23754.31	23792.26	26962.20	28821.00	30181.95
Highway	51374.74	59534.86	55738.08	56846.90	57955.72	61080.10	66771.52	71249.21	59636.39
Waterway	75423.84	81707.58	79435.65	92774.56	91772.45	97338.80	98611.25	99052.82	103963.04
Aviation	173.91	163.89	170.29	187.77	208.07	222.45	243.55	262.5	263.2
Total	156438.28	170593.42	164517.91	177339.42	173690.55	182433.61	192588.52	199385.53	194044.58

Table A-2
Energy consumption coefficient of various transport modes from 2011 to 2019 (Tonnes of standard coal per 100 million tonne-kilometres)

	2011	2012	2013	2014	2015	2016	2017	2018	2019
National railway	476	474	463	455	468	471	433	411	394
Highway professional freight company	2200	1700	1900	2000	1900	1740	1800	2000	1700
Ocean and coastal freight company	378	334.8	318.6	275.4	280.8	267.2	237.6	221.4	259.2
China civil aviation	4071.43	4185.71	4442.86	4342.86	4200.00	4185.71	4185.71	4100.00	4071.43

Table A-3
Situation of infrastructure and mobile equipment of various transport modes from 2011 to 2019

	2011	2012	2013	2014	2015	2016	2017	2018	2019
Operation mileage (Excluding high-speed railway) (10000 km)	86599	88244	92072	95344	101162	101020	101836	101796	104811
Locomotive (unit)	20721	20797	20835	21096	21366	21453	21420	21482	22000
Freight wagon (coach)	651175	670801	721850	716578	768516	764783	808736	839213	878000
Mileage (10000 km)	410.64	423.75	435.62	446.39	457.73	469.63	477.35	484.65	501.25
Highway commercial wagon (10000 units)	1179.4	1253.2	1419.5	1453.4	1389.2	1351.8	1368.6	1355.8	1087.8
Capacity of highway commercial wagon (10000 tonnes)	7261.2	8062.1	9614	10292.5	10366.5	10826.8	11774.8	12873	13587
Berths in ports above designated size (unit)	31968	31862	31760	31705	31259	30388	27578	23919	22893
Civil transport vessel (unit)	179000	179000	173000	172000	165900	160100	144900	137000	131600
Dead Weight Tonnage of civil transport vessel (10000 tonnes)	21264	22849	24401	25785	27244	26623	25652	25115	25685
Civil aviation route mileage (Calculated by unique distance) (10000 km)	349.06	328.01	410.6	463.72	531.7	634.8	748.3	837.98	948.22
Civil aviation airport (unit)	178	180	190	200	206	216	228	233	237
Civil transport aircraft (unit)	1764	1941	2145	2370	2650	2950	3296	3639	3818

Appendix B. Parameters determination method

B1 Situation of freight transportation in the base year

The freight transportation situation in the base year according to the statistical data is shown in Table B-1.

Table B-1
Freight turnover of various transport modes in the base year (2019)

transport modes	$Q_i(0)$(100 million tonne-kilometres)
Railway	30181.95
Highway	59636.39
Waterway	103963.04
Aviation	263.20
Total	194044.58

B2 Prediction of freight transportation demand in the target year

When studying the relationship between transportation and economic development, Rong found that the development of China's transportation sector gradually entered the 2.0 stage of transportation around 2010, in which the freight transportation demand was satisfied [35]. It can be assumed

that national freight transportation demand and actual freight turnover have been numerically equal since 2011. Therefore, based on the actual freight turnover of various transport modes from 2011 to 2019, we obtained the prediction results of freight transportation demand for various transport modes $Q_i^{\text{demand}}(n)$, as shown in Table B-2.

Table B-2
Freight transportation demand of various transport modes in the target year (2030)

transport modes	$Q_i^{\text{demand}}(n)$(100 million tonne-kilometres)
Railway	34349.36
Highway	70014.54
Waterway	111807.88
Aviation	285.19
Total	216465.97

B3 Prediction of freight transportation capacity in the target year

The freight transportation capacity of various transport modes is determined by factors such as the infrastructure and mobile equipment. Freight transportation capacity can be achieved only when infrastructure and mobile equipment are fully utilized to maximise technical efficiency [36].

In this study, a super-efficiency SBM-DEA model was applied from the output perspective to determine the technical efficiency of various transport modes. The infrastructure and mobile equipment of railways, highways, waterways, and aviation were considered as input indicators, and their freight turnover were considered as an output indicator. The SMB-DEA model is used to pursue the maximum of the technical efficiency of the various transport modes, i.e. to make full use of infrastructure and mobile devices. When the infrastructure and mobile devices (input indicators) are fully utilized, the volume of freight carried by the different transport modes in the target year (output indicators) can be used as a forecast of future carrying capacity.

The SBM-DEA model is able to analyse input redundancies and output shortages and thus improve technical efficiency accordingly. Based on the rationale of the SBM-DEA model, transport efficiency scores are high when the freight capacity of all modes of transport is fully utilized. In this state, the freight turnover of all modes of transport in the target year can be expressed as freight capacity.

There are many detailed descriptions of this method, such as those by Tone and Tian et al. [37,38]. The basic principle of the model is as follows.

There is a total of T decision units, denoted as $\text{DMU}_\theta(\theta = 1, 2\cdots, T)$. Each DMU has p types of input, denoted as $g_\tau(\tau = 1, 2, \cdots, p)$, and q types of output, denoted as $h_\omega(\omega = 1, 2, \cdots, q)$. $g_{\tau\theta}$ represents the input τ in the decision unit θ. $h_{\omega\theta}$ represents the output θ in the decision unit ω. λ_θ represents the weight of the decision unit θ.

$$\rho^* = \min \rho = \min \frac{\left(\frac{1}{p}\right)\sum_{\tau=1}^{p}\frac{s^-}{g_{\tau\theta}}}{\left(\frac{1}{q}\right)\sum_{\omega=1}^{q}\frac{s^+}{h_{\omega\theta}}} \text{ s.t.} \begin{cases} s_\tau^- \geq \sum_{\theta=1,\theta\neq\vartheta}^{T} g_{\tau\theta}\lambda_\theta \\ s_\omega^+ \leq \sum_{\theta=1,\theta\neq\vartheta}^{T} h_{\omega\theta}\lambda_\theta \\ \lambda_\theta \geq 0, s^- \geq g_{\tau\theta}, s_\omega^+ \leq h_{\omega\theta} \end{cases}$$

In the above formula, ρ^* is the technical efficiency of each DMU, and s_τ^- and s_ω^+ represent the slack of input and output variables, respectively. In this model, input redundancy and output shortage can be determined by s_τ^- and s_ω^+ to achieve an effective state.

Based on the national historical data from 2011 to 2019, the situation of infrastructure and mobile equipment (input indicators) of various transport modes in the target year was determined using the method of least squares curve fitting [39] (as shown in Table B-3).

Table B-3
Input indicators of various transport modes in the target year (2030)

Input indicators		Prediction results
Railway	Operation mileage (Excluding high-speed railway) (10000 km)	12.21
	Locomotive (unit)	24230.00
	Freight wagon (coach)	1305509.00
Highway	Mileage (10000 km)	523.69
	Highway commercial wagon (10000 units)	1508.30
	Capacity of highway commercial wagon (10000 tonnes)	13737.60
Waterway	Berths in ports above designated size (unit)	15470.00
	Civil transport vessel (unit)	86130.00
	Dead Weight Tonnage of civil transport vessel (10000 tonnes)	28758.00
Aviation	Civil aviation route mileage (Calculated by unique distance) (10000 km)	1183.48
	Civil aviation airport (unit)	259.00
	Civil transport aircraft (unit)	4711.00

Based on the prediction results of the input indicators, the freight turnover (output indicator) of different transport modes in the target year can be calculated when infrastructure and mobile equipment (input indicators) are fully utilized, that is, the prediction results of freight transportation capacity $Q_i^{\text{capacity}}(n)$ listed in Table B-4.

Table B-4
Freight transportation capacity of various transport modes in the target year (2030)

transport modes	$Q_i^{capacity}(n)$(100 million tonne-kilometres)
Railway	43582.63
Highway	76839.50
Waterway	119043.78
Aviation	281.39

B4 Prediction of freight transportation development speed and scale

(i) Prediction of development speed

Based on the economic and social development trend, Li reasonably predicted freight transportation demand in the next 30 years in China [40]. With the further deepening of economic transformation and supply side structural reforms, it is assumed that the gap between the proportions of tertiary and secondary industries will be further narrowed, or even overtaken, so that the growth rate of freight transportation demand will gradually decline. However, as industrialisation and urbanisation continue, the freight transportation demand will continue to grow at a low rate. Based on the analysis of the declining volatility trend in the annual growth rate of freight turnover of various transport modes from 2011 to 2019, we assume that the annual growth rate of total freight turnover $v = 1.5\%$.

Generally, combining the relatively stable positive correlation between economic growth and freight transportation development in the process of industrialisation with the continuous and steady growth of China's economy, the proportion of the transportation volume undertaken by various freight transport modes in the future is basically the same as the variation range in the past ten years [41]. Therefore, based on the variation range of the annual freight turnover growth rate of various transport modes from 2011 to 2019, the variation range of the annual freight turnover growth rates of railways, highways, waterways, and aviation from 2021 to 2030 were set ($[m_i, M_i]$ as shown in Table B-5).

Table B-5
Variation range of the annual freight turnover growth rate of various transport modes

transport modes	m_i	M_i
Railway	0.2%	5.8%
Highway	0.4%	6.7%
Waterway	0.4%	7.1%
Aviation	0.1%	7.8%

(ii) Prediction of development scale

In this study, combined with the development status and requirements of various transport modes, the regulation range of the share rate of freight turnover of various transport modes $[c_i, C_i]$ is predicted as follows:

For railway transportation, the share rate of railway freight turnover was 15.55% in 2019, and this value fluctuated between 13% and 20% over the recent years. Railway transportation has the characteristics of low energy consumption and strong transportation capacity and has obvious comparative advantages over highway transportation in bulk cargo and mid-long-distance transportation. Therefore, China has formulated relevant incentive policies and mandatory goals to expand the scale of railway transportation development. In this study, the regulation range of the share rate of railway freight turnover was set to $[c_1, C_1] = [15.55\%, 20\%]$.

For highway transportation, over the recent years, the share of highway freight turnover has fluctuated between 30% and 36%. Owing to the high energy consumption of highway transportation, the high share rate of highway freight turnover is not conducive to the realisation of energy saving and emission reduction goals. Therefore, against the background of vigorously advocating the 'highway-to-railway shift' and 'highway-to-waterway shift' China has proposed a requirement to limit the highway transportation development scale. In this study, the regulation range of the share rate of highway freight turnover was set as $[c_2, C_2] = [30\%, 36\%]$.

For waterway transportation, from 2011 to 2019, the share rate of waterway freight turnover continued to rise by approximately 5% points, and as of 2019, this value reached 53.58%. The waterway is the transport mode with the lowest unit energy consumption and largest transportation capacity. The vigorous development of coastal and inland shipping will positively impact promoting the construction of green transportation systems. Therefore, in this study, the regulation range of the share rate of waterway freight turnover was set as $[c_3, C_3] = [50\%, 58\%]$.

For aviation transportation: With the rapid development of e-commerce logistics, the share rate of aviation freight turnover has steadily increased in recent years to satisfy the high-quality social transportation demand, and this value has exceeded 0.12% in 2019. However, considering the goal of energy conservation and emission reduction, it is necessary to control the growth rate of its share in the future because aviation is the transport mode with the largest unit energy consumption. Therefore, in this study, the regulation range of the share rate of aviation freight turnover was set as $[c_4, C_4] = [0.12\%, 0.15\%]$.

The lower and upper limits of the freight transportation development scale of various transport modes were calculated using the following formula, and the results are listed in Table B-6.

$$\begin{cases} Q_i^{\text{lower}}(n) = c_i \sum_{i=1}^{r} Q_i^{\text{demand}}(n) \\ Q_i^{\text{upper}}(n) = c_i \sum_{i=1}^{r} Q_i^{\text{demand}}(n) \end{cases}$$

Table B-6
Freight transportation development scale of various transport modes in the target year (2030)

transport modes	$Q_i^{\text{lower}}(n)$(100 million tonne-kilometres)	$Q_i^{\text{upper}}(n)$(100 million tonne-kilometres)
Railway	33659.06	43291.39
Highway	64937.09	77924.51
Waterway	108228.49	125545.04
Aviation	259.74	324.69

B5 Prediction of freight transportation energy consumption in the target year

Based on the historical data of national railways, professional highway-based freight companies, ocean and coastal freight companies, and China civil aviation from 2011 to 2019, we predicted the energy consumption coefficients of various transport modes $e_i(n)$ by the Gray Prediction model. We use the most widely used GM (1,1) model. The basic principle of the model is as follows [42].

There are more than ten different forms of the GM (1,1) model. We take the form of a whitened differential equation as shown below.

$$\frac{dH^{(1)}}{dw} + aH^{(1)} = b$$

In the above formula, $\frac{dH^{(1)}}{dw}$ is the gray derivative, $H^{(1)}$ is the background value, a is the development coefficient, and b is the gray action amount. The prediction results are shown in Table B-7.

Table B-7
Energy consumption coefficients of various transport modes in the target year (2030)

transport modes	$e_i(n)$(Tonnes of standard coal per 100 million tonne-kilometres)
Railway	328.00
Highway	1540.00
Waterway	199.40
Aviation	3903.40

To control the energy consumption of freight transportation, the value of the energy consumed when ensuring that the freight transportation demand of various transport modes can be satisfied is considered as the expected value of the total energy consumption of freight transportation, that is, $E = \sum_{i=1}^{r} Q_i^{\text{demand}}(n) e_i(n)$.

The expected value of total freight transportation energy consumption was 145 million tonnes of standard coal in the target year.

References

[1] Chen J. High-speed rail and energy consumption in China: the intermediary roles of industry and technology. Energy 2021;230:120816.
[2] Choi Y, Zhang N, Zhou P. Efficiency and abatement costs of energy-related CO_2 emissions in China: a slacks-based efficiency measure. Appl Energy 2012;98:198–208.
[3] Chang Y, Zhang N, Danao D, et al. Environmental efficiency analysis of transportation system in China: a non-radial DEA approach. Energy Pol 2013;58:277–83.
[4] Stefaniec A, Hosseini K, Xie J, et al. Sustainability assessment of inland transportation in China: a triple bottom line-based network DEA approach. Transport Res Transport Environ 2020;80:102258.
[5] Zhu Q, Li X, Li F, et al. Energy and environmental efficiency of China's transportation sectors under the constraints of energy consumption and environmental pollution. Energy Econ 2020;89:1–9.
[6] Zhang Y, Peng H, Liu Z, et al. Direct energy rebound effect for road passenger transport in China: a dynamic panel quantile regression approach. Energy Pol 2015;87:303–13.
[7] U.S. Department of Transportation. National freight strategic plan. Available at: https://www.transportation.gov/freight/NFSP; 2020 (2020), Accessed 25 May 2021.
[8] Department UK. Of transport. Transport decarbonization plan. 2021 [Available at: Department for Transport - GOV.UK (www.gov.uk)].
[9] State Council of the People's Republic of China (SCPRC). Three-year action plan for promoting the adjustment of transportation structure (2018-2020). 2018. Available at: http://www.gov.cn/zhengce/content/2018-10/09/content_5328817.htm.
[10] Churchman P, Longhurst N. Where is our delivery? The political and socio-technical roadblocks to decarbonising United Kingdom road freight. Energy Res Social Sci 2022;83:102330.
[11] European Commission: Roadmap to a Single European Transport Area – Towards a competitive and resource efficient transport system 2011. EUR-Lex - 52011DC0144 - EN - EUR-Lex (europa.eu).
[12] International Energy Agency (IEA). The future of rail-Opportunities for energy and the environment. Available at: https://www.iea.org/terms; 2019.
[13] Goss WP, Mcgowan JG. Transportation and energy — a future confrontation. Transportation 1972;1(3):265–89.
[14] Neves SA, Marques AC. The substitution of fossil fuels in the US transportation energy mix: are emissions decoupling from economic growth? Res Transport Econ 2021:101036.
[15] Batur İ, Bayram IS, Koc M. Impact assessment of supply-side and demand-side policies on energy consumption and CO_2 emissions from urban passenger transportation: the case of Istanbul. J Clean Prod 2019;219:391–410.
[16] Yu Y, Li S, Sun H, et al. Energy carbon emission reduction of China's transportation sector: an input–output approach. Econ Anal Pol 2021;69:378–93.
[17] Umar M, Ji X, Kirikkaleli D, et al. The imperativeness of environmental quality in the United States transportation sector amidst biomass-fossil energy consumption and growth. J Clean Prod 2021;285:124863.
[18] Shi R, Hu Z, Zhou Y, et al. Research on railway freight market share based on the whole process of transport. Procedia-Soc Behav Sci 2014;138:298–304.
[19] Xiang Y, Wang W, Zheng D, et al. Dominant transportation distance for multi transport modes in regional integrated freight network. J Transport Syst Eng Info Technol 2016;16(6):33–9.
[20] Dong B, Wu Q, Peng Z. Analysis on evolution characteristics and tendency and integrated transport system structure. J Highw Transp Res Dev 2017;34(6):151–8.
[21] Lee M, Yoo S. The role of transportation sectors in the Korean national economy: an input-output analysis. Transport Res Pol Pract 2016;93:13–22.

[22] Cai J, Liu Y, Zhang M, et al. A study on influencing factors and strategies of transport structure adjustment in Beijing. Railway Transp Econ 2019;41(6):59–63.
[23] Cui Q, He L, Liu Y, et al. The impacts of COVID-19 pandemic on China's transport sectors based on the CGE model coupled with a decomposition analysis approach. Transport Pol 2021;103:103–15.
[24] Carrier RE. Energy conservation through urban transportation planning. Transport Res 1974;8(4–5):493–501.
[25] Komor P. Reducing energy use in US freight transport. Transport Pol 1995;2(2):119–28.
[26] Álvarez AG. Energy consumption and emissions of high-speed trains. Transport Res Rec 2010;2159:27–35.
[27] Tian Y, Zhu Q, Lai K, et al. Analysis of greenhouse gas emissions of freight transport sector in China. J Transport Geogr 2014;40:43–52.
[28] Wang B, Sun Y, Chen Q, et al. Determinants analysis of carbon dioxide emissions in passenger and freight transportation sectors in China. Struct Change Econ Dynam 2018;47:127–32.
[29] Ülengin F, Işık M, Ekici ŞÖ, et al. Policy developments for the reduction of climate change impacts by the transportation sector. Transport Pol 2018;61:36–50.
[30] Li L, Zhang X. Reducing CO_2 emissions through pricing, planning, and subsidizing rail freight. Transport Res Transport Environ 2020;87:102483.
[31] Tang J, Wang C, Zuo D. Optimization model and arithmetic of region transportation structure based on optimization adjusting path. Mathematics in Practice and Theory 2013;43(13):85–91.
[32] Saaty TL. How to make a decision: the analytic hierarchy process. Eur J Oper Res 1990;48(1):9–26.
[33] Murty KG, Fathi Y. A feasible direction method for linear programming. Oper Res Lett 1984;3(3):121–7.
[34] Heinold A, Meisel F. Emission rates of intermodal rail/road and road-only transportation in Europe: a comprehensive simulation study. Transport Res Transport Environ 2018;65:421–37.
[35] Rong C. On the development stage of transport industry and its new normal and supply side reform. China Transport Rev 2016;38(12):1–6+10.
[36] Wu Q, Song J, Ju P, et al. Spatial distribution of provincial integrated transport efficiency in China. Econ Geogr 2015;35(12):43–9.
[37] Tone K. A slacks-based measure of super-efficiency in data envelopment analysis. Eur J Oper Res 2002;134(1):32–41.
[38] Tian N, Tang S, Che A, et al. Measuring regional transport sustainability using super-efficiency SBM-DEA with weighting preference. J Clean Prod 2020;242:118474.
[39] Park J, Seo Y, Ha M. The role of maritime, land, and air transportation in economic growth: panel evidence from OECD and non-OECD countries. Res Transport Econ 2019;78:100765.
[40] Li Q. China's freight demand trend in the coming 30 years. China Transport Rev 2020;42(10):101–5.
[41] Wang H, Han J, Su M, et al. The relationship between freight transport and economic development: a case study of China. Research in Transportation Economics; 2020, 100885.
[42] Deng J. Control problems of grey systems. Syst Control Lett 1982;1(5):288–94.